

Suggested Teaching Guidelines for
Data Visualization - Analysis and Reporting
PGCP-BDA February 2026

Duration: 26 Hours Theory + 24 Hours Lab + 40 Hours Self Learning Objective: To introduce students in Data Analytics, Visualization and Reporting

Prerequisites: Knowledge of Database Fundamentals and Big Data Technologies.

Evaluation method: Theory exam– 40% weightage Lab exam – 40% weightage
Internal exam– 20% weightage

List of Books / Other training material

Courseware: No specific courseware for the module, faculty may share somecoursematerials.**Reference Book:**

1. Communicating Data with Tableau, Ben Jones, O'Reilly, Shroff Publishers&Distributors,Tableau 8.1.
2. Mastering Microsoft Power BI: Expert Techniques for Effective DataAnalyticsand Business Intelligence Book by Brett Powell
3. Designing Data Visualizations, by Steele,O'Reilly
4. Tableau your data, by Daniel G/ Wiley
5. Graphs Cookbook, Hrishi V. Mittal, Packt Publishing 6. Python Data Visualization Cookbook,Igor Milovanović, Packt Publishing7. Learning Python Data Visualization, Chad Adams, Packt Publishing
8. Data Visualization with D3.js Cookbook,Nick Qui Zhu,Packt Publishing9. Getting Started with D3,Mike Dewar,O'Reilly
10. Data Visualization with JavaScript
11. Data Visualization for Dummies
12. High Impact Data Visualization with Power View, Power Map, andPower BI 13. The Visual Organization: Data Visualization, Big Data, and the Quest for BetterDecisions
14. Mastering Tableau 2021:- by Marleen Meier

Session 1 & 2: (4T +2L+4SL)
Theory and Lab:

- Business Intelligence basics
- Information gathering
- Decision making
- Managing BI
- BI User Segmentation
- Gathering BI Requirements
- Content and Knowledge Management

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- Strategic Approach to BI
- Significance of visual analytics Information Visualization ○ Data Representation
- Data collection and binding
 - Structured Data
 - Unstructured data

Self Learning:

- Coffee Chain Case Study (*Note: Self Learning in further sessions is based on this coffee chain case study*)
- Identify Stakeholders for coffee chain:
 - Executives (CEO, CFO) - Strategic decisions
 - Regional Managers - Regional performance
 - Store Managers - Daily operations
 - Marketing Team - Customer insights
 - Inventory Manager - Supply chain
- Define BI Requirements by user segment:
 - List 3-4 key questions each stakeholder needs answered
 - Identify KPIs for each role (revenue, profit margin, footfall, inventory turnover, customer satisfaction)
 - Determine reporting frequency (daily, weekly, monthly)
 - Define data needs (sales, inventory, customer, product, store)
- Create BI Requirements Document:
 - Stakeholder matrix with information needs
 - KPI definitions and calculation formulas
 - Data source identification
 - Success metrics for BI implementation

Session 3, 4 & 5: (6T+6L+8SL)

Theory and Lab:

MS EXCEL

- Functions
- Formula
- Charts
- Pivots and Lookups
- Data Analysis Tool pack
 1. Descriptive Summaries
 2. Correlation
 3. Regression

Self Learning:

- Advanced Functions

- Explore array formulas and dynamic arrays (FILTER, SORT, UNIQUE,

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- Master advanced lookup functions: XLOOKUP with multiple criteria, INDEX-MATCH-MATCH for 2D lookups
- Learn SUMPRODUCT for complex conditional calculations ■ Practice nested IF function
- Implement error handling with IFERROR, IFNA, ISERROR
- Advanced Statistical Analysis
 - Explore additional Data Analysis ToolPak features: ANOVA, t-tests, F-tests ■ Understand confidence intervals and hypothesis testing ■ Practice moving averages and exponential smoothing for forecasting ■ Create correlation matrices with heatmap visualization ■ Implement multiple regression with residual analysis

Self Learning:

- Dataset: Coffee chain sales data with tables:
 - Sales Transactions: Date, Store_ID, Product_ID, Quantity, Unit_Price, Discount, Total_Revenue
 - Products: Product_ID, Product_Name, Category, Cost_Price, Size
 - Stores: Store_ID, Store_Name, Location, City, State, Manager, Opening_Date
 - Customers: Customer_ID, Visit_Date, Loyalty_Tier, Purchase_Amount
- Data Preparation:
 - Import all datasets into Excel
 - Use VLOOKUP/XLOOKUP to combine data from multiple sheets ■ Create calculated columns: Profit = Revenue - (Quantity × Cost_Price), Profit_Margin = Profit/Revenue
- Descriptive Statistics:
 - Use Data Analysis ToolPak for descriptive statistics on Revenue, Profit, Quantity
 - Calculate mean, median, mode, standard deviation by Store and Product Category
 - Identify top/bottom performers using RANK function
- Advanced Formulas:
 - SUMIFS: Total revenue by store, category, date range ■ AVERAGEIFS: Average transaction value by customer tier ■ COUNTIFS: Number of transactions by product category ■ Nested IF: Categorize stores as "High", "Medium", "Low" performers based on revenue thresholds
 - Date functions: MONTH, YEAR, WEEKDAY to extract temporal patterns
- Correlation Analysis:
 - Use Data Analysis ToolPak to find correlations between:
 - Store Size vs Revenue
 - Discount % vs Profit Margin
 - Customer Loyalty Tier vs Average Purchase Amount
 - Create correlation matrix and interpret results

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- Regression Analysis:
 - Build regression model predicting Monthly Revenue based on: ■ Store Location (urban/suburban)
 - Average Daily Footfall
 - Product Mix (% beverages vs food)
 - Average Discount %
 - Interpret R-squared, coefficients, significance
- Pivot Tables:
 - Create pivot tables:
 - Revenue by Store and Month with growth %
 - Product Category performance with profit margins
 - Customer segments by loyalty tier
 - Add slicers for Date, Store, Category

Session 6: (2T+2L+4SL)

Theory and Lab:

Data analytics Life Cycle

- Discovery
- Data preparation
- Model planning
- Model building implementation
- Quality assurance
- Documentation
- Management approval
- Installation
- Acceptance and operation

Self Learning:

- Discovery Phase:
 - Document 5 key business questions: "Which products are most profitable?", "What factors drive store performance?", "How to optimize inventory?", etc. ■ Identify available data sources
 - Define success criteria and metrics
- Data Preparation:
 - Assess data quality: missing values, outliers, inconsistencies ■ Document data cleaning requirements
 - Plan data integration strategy
- Model Planning:
 - Select analytical approaches for each business question ■ Choose appropriate visualization types

- Plan Excel vs Tableau usage
- Communication Strategy:
 - Design dashboard layout mockups on paper
 - Plan presentation structure for different stakeholders

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- Define refresh schedule and maintenance plan

Session 7 & 8: (4T+4L+8SL)
Theory and Lab:

- Introduction to Tableau
- Intelligent data analysis
- Nature of Data
- Analytics Processes and tools
- Analysis vs. Reporting
- Modern Data Analytic Tools
- Data sources in Tableau

Self Learning:

- Compare major BI tools: Tableau, Power BI
- Identify when to use Excel vs Tableau for different scenarios ○ Explore emerging analytics tools: Python libraries (Matplotlib, Seaborn, Plotly), R(ggplot2)
- Understand cloud BI platforms vs on-premise solutions ○ Study real-time analytics vs batch processing requirements ○ Understanding Which tool must be used for which task ○ *Case Study:*
- Tableau Setup:
 - Connect to coffee chain Excel file or CSV ■ Understand Tableau interface: Data pane, Sheets, Dashboards ■ Explore data source: Dimensions vs Measures, data types ○
- Basic Visualizations:
 - Create bar chart: Revenue by Store
 - Create line chart: Daily sales trend
 - Create scatter plot: Store Size vs Revenue
 - Apply basic filters and sorting
- Analysis vs Reporting:
 - Reporting Example: Create standardized weekly sales report showing fixed metrics
 - Analysis Example: Explore why Store X underperforms - drill down into products, time periods, customer segments
 - Document differences in approach and output
- Data Source Management:
 - Connect to multiple data sources (Excel sheets)
 - Create relationships between Sales, Products, Stores tables ■ Understand live vs extract connections
 - Create custom data source with proper relationships

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Session 9, 10 & 11: (6T+6L+8SL)

Theory and Lab:

- Visualization Algorithms
- Visual Encodings
 - color, size, shape, lines, axes, scaling, annotation
 - Taxonomy of data visualization (Some Types of charts, but not limited to)
 - Comparison charts – types of Bar chart, Box plots, Histograms, Gantt charts, Bullet graphs, side-by-side bar chart etc.
 - Tables – Text Tables, Highlight tables
 - Hierarchies and relationships – Pie chart, stacked bar, Tree map etc.
 - Changes over time – Line chart, dual lines, Area charts etc.
 - Connections and relationships – scatter plots, Symbol maps, map, heat maps, packed bubble chart etc.

Self Learning:

- Build different charts:
- Comparison Charts:
 - Horizontal Bar: Top 10 products by revenue
 - Side-by-side Bar: Revenue comparison across stores for different product categories
 - Bullet Graph: Store performance vs target revenue
 - Box Plot: Distribution of transaction values by store location
- Tables:
 - Text Table: Store-wise monthly revenue grid
 - Highlight Table: Heat table showing profit margins by Product × Store
- Hierarchies & Composition:
 - Treemap: Revenue breakdown by Category → Product
 - Stacked Bar: Product mix (beverages vs food) across stores
- Pie Chart: Revenue contribution by top 5 product categories
- Time Series:
 - Line Chart: Daily sales trend over 3 months with trend line
 - Dual-Axis Line: Revenue (line) and Profit Margin % (line) over time
 - Area Chart: Stacked area showing category contribution over time
- Relationships & Distribution:
 - Scatter Plot: Discount % vs Profit Margin with trend line and R-squared
 - Bubble Chart: Stores positioned by Revenue (X) vs Profit Margin (Y), bubblesize = Customer Count
 - Histogram: Distribution of transaction values
 - Heat Map: Sales intensity by Day of Week × Hour of Day (if time data available)

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Session 12 &13: (4T+4L+8SL)

Theory:

- Choosing appropriate visuals
- Applying calculations using functions, statistics
 - Numeric Calculations
 - String Calculations
 - Date calculations
 - LOD (Level of Detail) Expressions
- Data sorting, filters
- Interactive visualization
 - Data updating
 - Visual updating
- Dashboard Design

Lab:

- Load coffee chain dataset in Tableau and create required visuals ■ Also create the dashboard for the same dataset using Tableau.

Self Learning:

- Create Calculated Fields
- Build 10-12 calculated fields covering:
 - Numeric: Profit, Profit Margin %, YoY Growth %, Running Total
 - String: Store labels, Product size categories, Performance ratings
 - Date: Days since opening, Quarter labels, Month names, Weekend/Weekdayflags
 - LOD: Customer lifetime value (FIXED), Store averages (INCLUDE), Category comparisons (EXCLUDE), Store rankings
- Build 3 Interactive Dashboards:
 - Dashboard 1 - Executive Overview:
 - High-level KPIs (revenue, profit, margin, store count), monthly trends, geographic map of stores, top 5 products/stores. Filters: date range, region.
 - Dashboard 2 - Store Performance:
 - Store revenue comparison, performance scatter plot (revenue vs margin with quadrants), product mix per store, growth trends. Interactive: click store to filter, parameter to switch metrics (revenue/profit/customers).
 - Dashboard 3 - Product Analytics:
 - Category treemap, profitability quadrant (volume vs margin), seasonal

heatmap, product relationships. Interactive: drill-down from category to products, filter and highlight actions.

- Add Interactivity:

- Implement dashboard actions (filter on click, highlight on hover), create parameters (metric selector, date comparison), add filters (date, store, category), design custom tooltips with additional context.